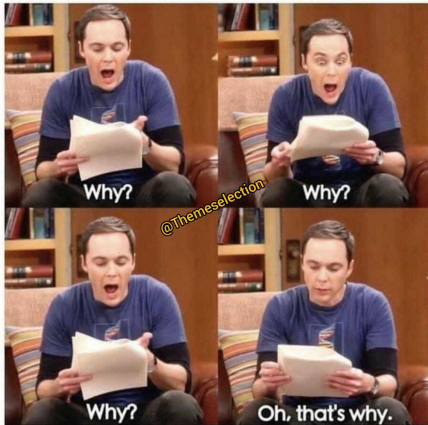


COGS 108 Week 6

Programmers while reviewing the codes

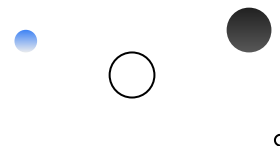


Adapted from slides in COGS 108 FA23 Week 7



LOGISTICS

DUE DATES



Sun Feb 23, 2025

 Checkpoint #1: Data

due by 11:59pm

 D6

due by 11:59pm

 Q7

due by 11:59pm





DISCUSSION

LAB 6

INFERENCEAL ANALYSIS



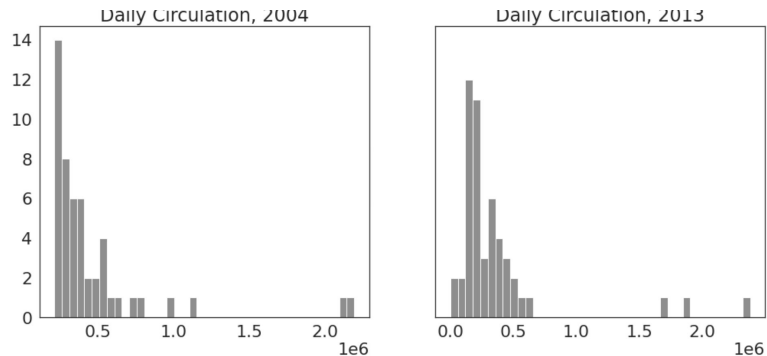


Get rid of the commas in the numbers for Daily Circulation:

```
df['Series'].str.replace(str).astype(float)
```

#Look at daily circulation distribution in 2004 and in 2013

Plot using sns.histplot(). Parameters used for plot below: bins=40.





Let's look at the distribution of Pulitzer prize winners for the same time period.

Plot using `sns.histplot()`

Parameters used for plot below: `bins=30`.

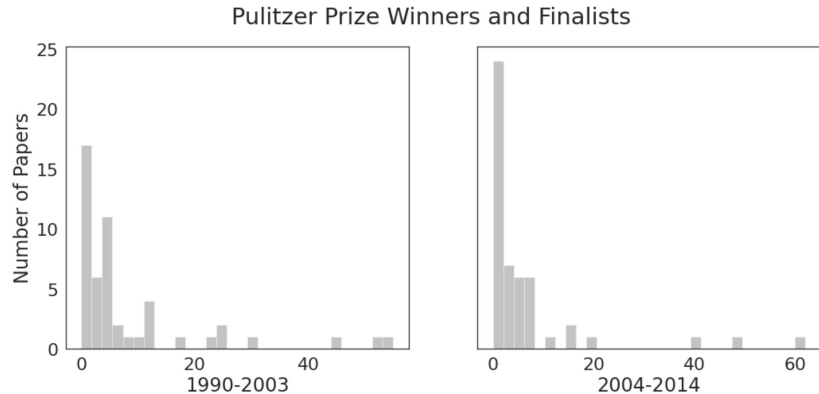
```
fig, (ax1, ax2) = plt.subplots(ncols=2, sharey=True)
```

```
sns.histplot(...)
```

```
ax1.set_title(...)
```

```
sns.histplot(...)
```

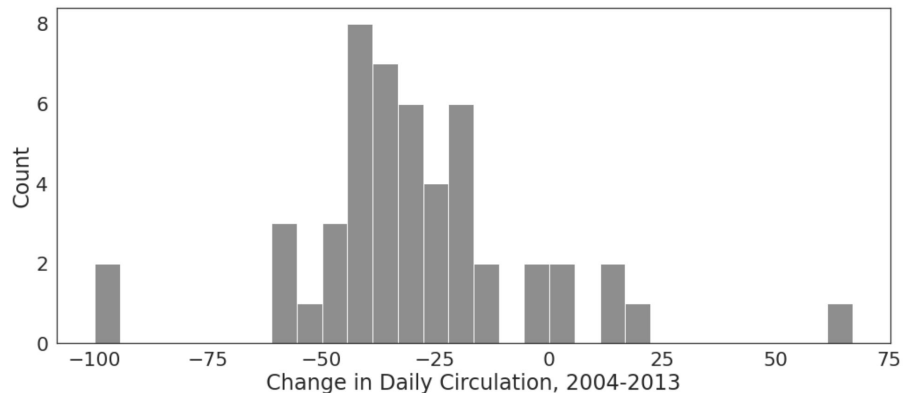
```
ax2.set_title(...)
```





Plot the distribution of "change" in daily circulation:

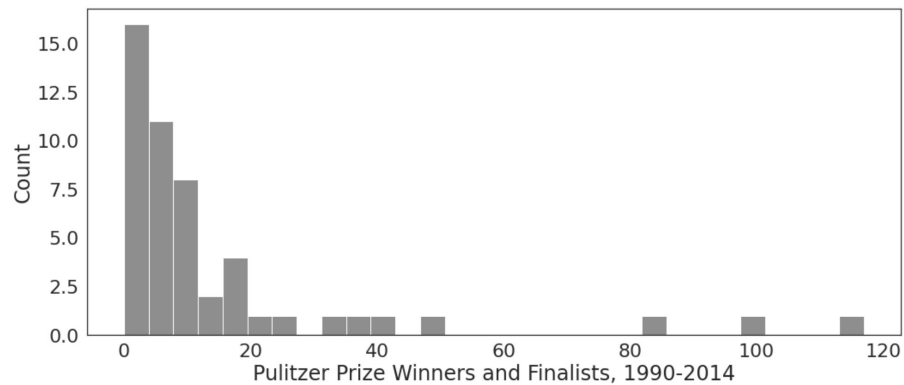
Plot using `sns.histplot()`,
parameters used for plot below: `kde=False`, `bins=30`, `color="dimgrey"`





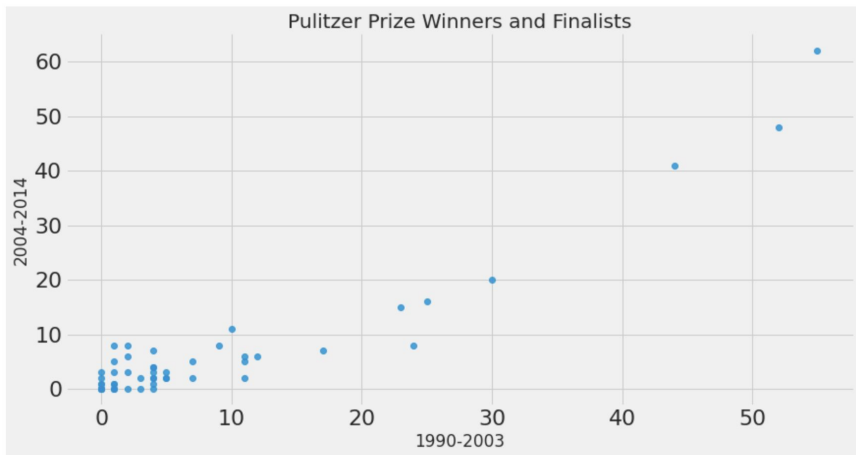
Look at pulitzer prize winner distributions:

Plot using sns.histplot()





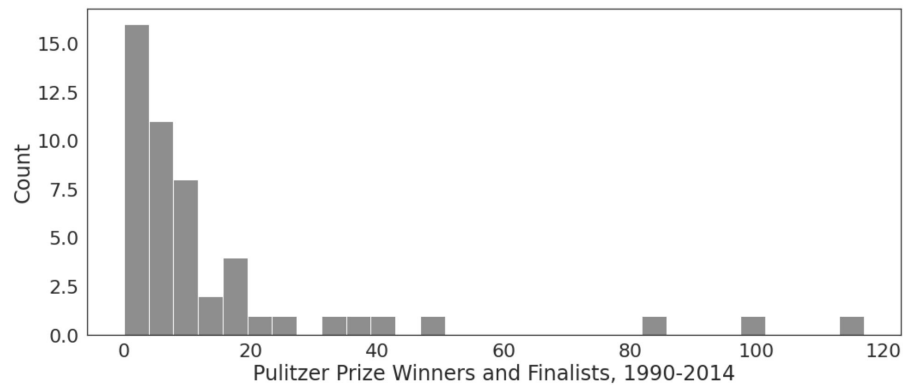
plot relationship between pulitzer prize winners/finalists in each time period and look at number of pulitzers between two time periods
Plot using `sns.Implot(x = 'Series1', y = 'Series2', data = DataframeName, fit_reg = False, height = 6, aspect=2)`





In the cell below look at pulitzer prize winner distributions

Plot using sns.histplot()



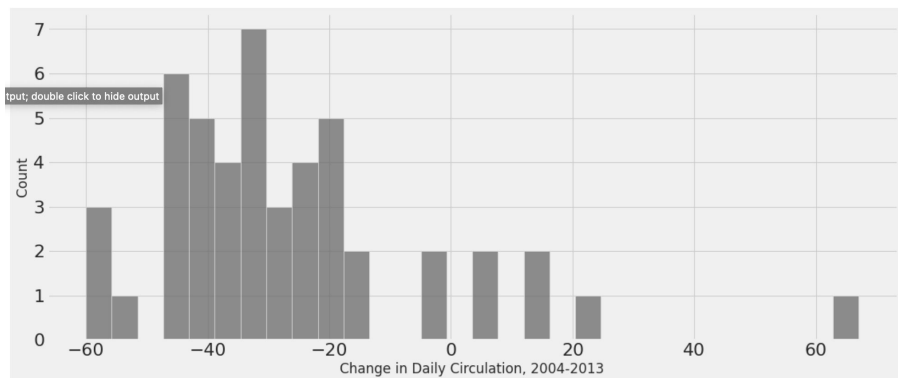


#Who has won the most pulitzers during the years we're looking at?

Use sort_values() to look at the top values

Parameter: ascending = False

#Plot the distribution of daily change in circulation after outlier removal

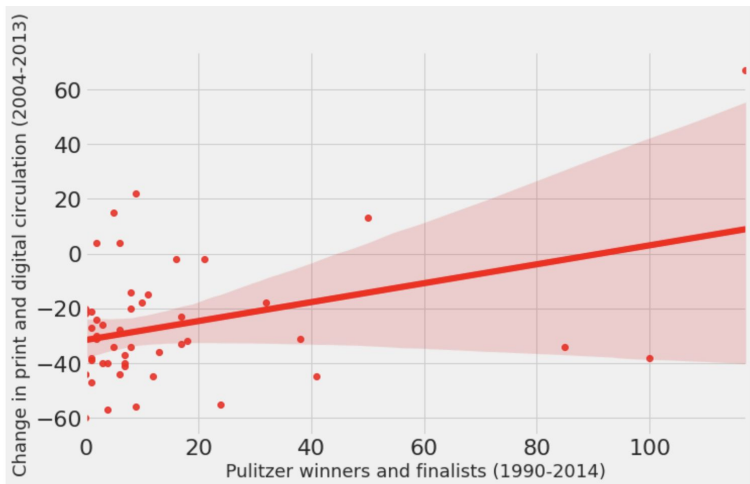




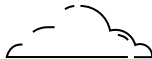
#Relationship between the total number of Pulitzers and change in readership
(daily circulation)

Use `sns.lmplot(x = 'Series1', y = 'Series2', data = dataFrameName, fit_reg = True,
height = 6, aspect = 1.7, line_kws={'color': 'red'}, scatter_kws={'color': 'red'})`

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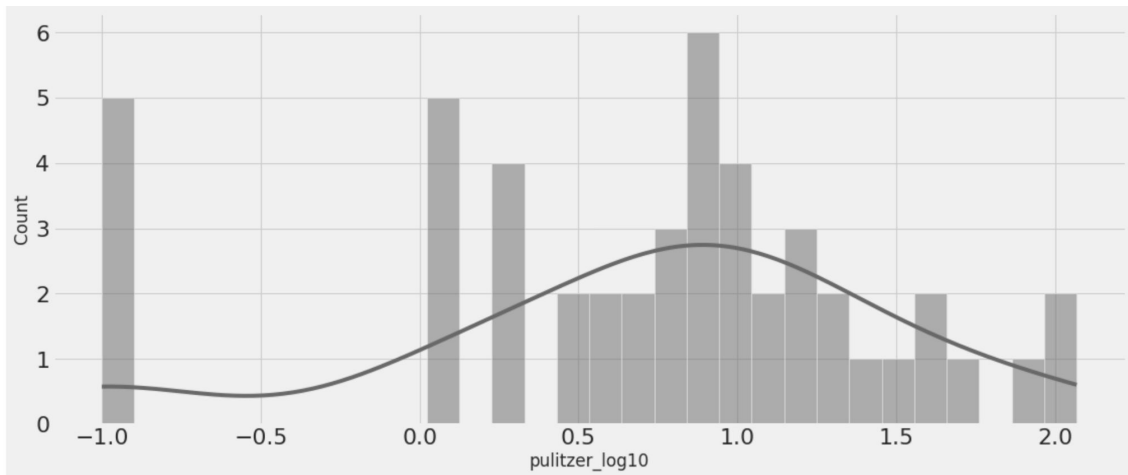


#Apply a log10-transformation the Pulitzer count data, with an offset of 0.1

```
Use Pulitzer['pulitzer_log10'] = np.log10 ( Series +0.1)
```

#In the next cell, visualize the distribution of the log10 column

Use sns.histplot()



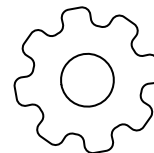
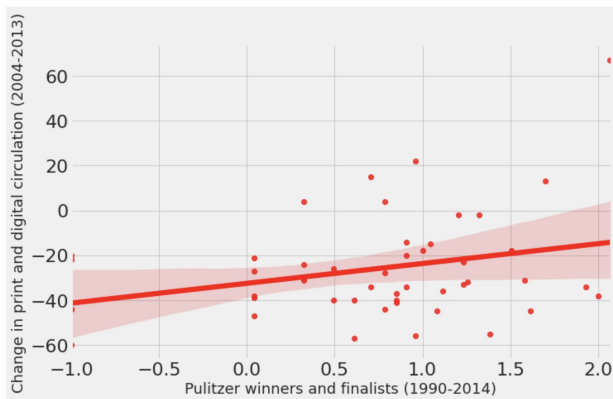
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#plot the relationship between our two variables of interest
Use `sns.lmplot()`





#Carry out linear regression; Now use statsmodels to initialize an OLS linear model This step initializes the model, and provides the data (but does not actually compute the model); fit the model; and Check out the results.

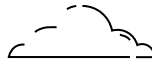
```
df = pulitzer[['Change in Daily Circulation, 2004-2013',  
              'pulitzer_log10']]  
df.columns = ['circulation', 'pulitzer_log10']  
df.head()
```

```
outcome, predictors = patsy.dmatrices('circulation ~ pulitzer_log10', df)
```

```
# Now use statsmodels to initialize an OLS linear model  
# This step initializes the model, and provides the data (but does not actually compute the model)  
mod_log = sm.OLS(outcome, predictors)
```

```
# fit the model  
res_log = mod_log.fit()
```

```
# Check out the results  
print(res_log.summary())
```





#Carry out linear regression => (A modern way)
Fit the model; and Check out the results.

```
df = pulitzer[['Change in Daily Circulation, 2004-2013',  
              'pulitzer_log10']]  
df.columns = ['circulation', 'pulitzer_log10']  
df.head()
```

We can use statsmodels.formula.api to run this linear regression without patsy

Go back up to the first cell and 'import statsmodels.formula.api as smf'

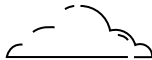
This step initializes the model, and provides the data (but does not actually compute the model)
mod_log = smf.ols(formula='circulation ~ pulitzer_log10', data=df)

fit the model
res_log = mod_log.fit()

Check out the results
print(res_log.summary())

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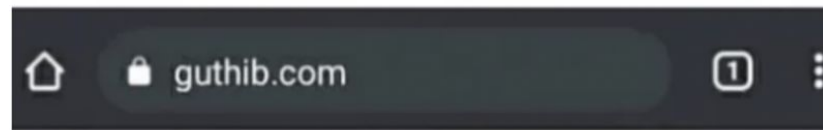


THANKS!

Questions on Edstem or office hours

CREDITS: This presentation template was created by **Slidesgo**, including icons by **Flaticon** and infographics & images by **Freepik**

Someone literally bought a domain to do this



You spelled it wrong.

